

Erdős Problem #951

EP-951

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $1 < a_1 < \dots$ be a sequence of real numbers such that $\left| \prod_i a_i^{k_i} - \prod_j a_j^{\ell_j} \right| \geq 1$ for every distinct pair of non-negative finitely supported integer tuples $k_i, \ell_j \geq 0$. Is it true that $\#\{a_i \leq x\} \leq \pi(x)$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>